

Prospecta

Long-term memory for beings and agents: retain what happened, recall what matters. How it works, technique by technique, with the numbers.

An explainer for a reader new to memory systems

From what it is to how it is measured

1

What it is

Memory for beings and agents, one bank each, on Postgres.

2

How it works

What is written for every note, and the plan for reading.

3

Every technique

Search, filters, graph, fusion, reranking, answers, sets.

4

Scoring

How a score is built: hit@1, hit@10, cover@10.

5

Four questions

A lookup, a paraphrase, a dated question, a set.

6

Why hybrid wins

The ladder from 0.72 to 0.99.

7

Tradeoffs

Cost, latency, write-time cost, hard lessons.

8

Improvements

What is still weak, said plainly.

9

Plugging in

Rung and Hermes, through thin providers.

Words used in this deck

Note

One entry in a memory: a diary page, a message, a document. Notes are stored whole, never rewritten.

Chunk

A piece of a note, at most 1,000 characters. Search works on chunks; answers cite the parent note.

Embedding

A list of 1,536 numbers that places a text on a map of meaning. Close points mean similar texts.

Bank

One being's private memory. Banks live side by side in one database and never mix.

Retain, recall

Retain writes a note in. Recall brings back what matters for a question.

LLM

A large language model, such as Sonnet. It reads and writes text; it costs money and seconds.

RUNNING EXAMPLE one invented diary, kept by Ines, a gardener, traveller and cook. No real note appears anywhere.

One invented diary runs through every example

Ines does not exist. Every note below is made up; the charts later are real measurements.

seed-order.md 18 Apr 2025 Ordered twelve seed packets from Fenwick Seeds: Black Krim tomatoes, Genovese basil, kale and nine more.	tomato-planting.md 12 May 2025 Planted the Black Krim tomatoes in bed 3. Soil was 16 degrees by Mr. Okafor's thermometer.	lisbon-stay.md 2 Jun 2025 A short stay at the Pensão Azul in Lisbon. Best breakfast: custard tarts on Rua Augusta.
fence-rebuild.md 21 Jun 2025 Mr. Okafor and I rebuilt the back fence. He brought cedar posts; I made lemon rice.	yellow-leaves.md 8 Jul 2025 Lower tomato leaves yellow with brown spots. Mr. Okafor says early blight. Sprayed copper.	heat-wave.md 19 Jul 2025 37 degrees. Watered twice. The basil bolted.
first-harvest.md 14 Aug 2025 First big harvest: nine kilos of Black Krim. Sauce: garlic, basil, red wine.	compost-skipper.md 2 Mar 2025 Dara turned the compost heap again. Family calls her Dari; I call her Skipper Compost.	zine-column.md 10 Sep 2025 Dara's first garden zine column ran under the pen name Fern Aldous.

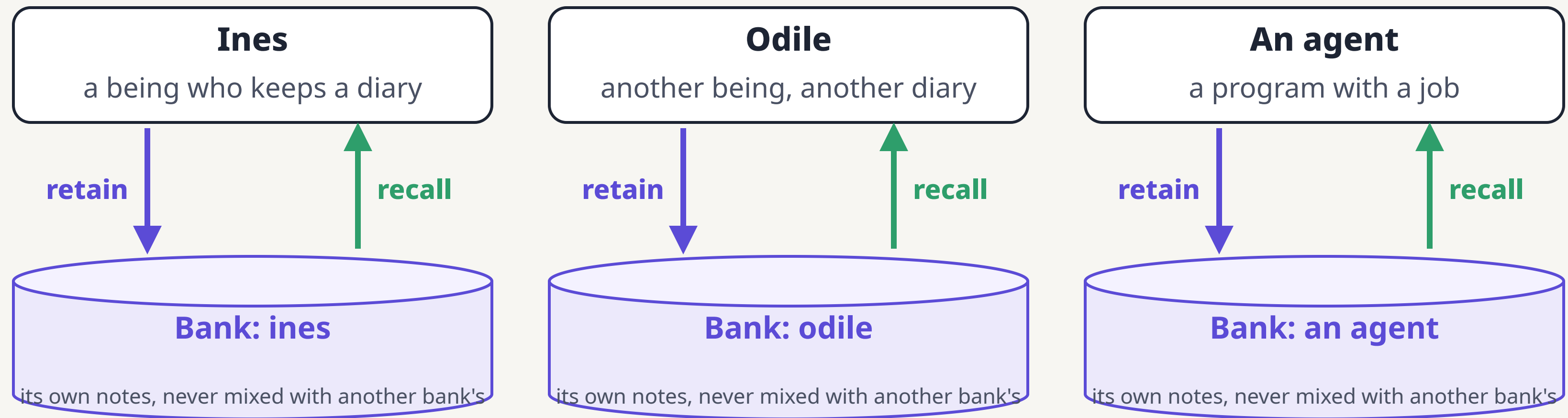
What Prospecta is

Long-term memory for beings and agents: what it keeps, where, and why it exists.

WHAT IT IS

Memory for beings and agents

Retain what happened, recall what matters, one bank per being.



Retain: write down what happened. Recall: bring back what matters to the question asked.

IN SHORT A being retains notes into its own bank and later recalls what matters for the question it faces.

How text becomes something searchable

A vector is a note's place on a map of meaning, stored in Postgres with pgvector.

A note's text

Planted the Black Krim tomatoes in bed 3.

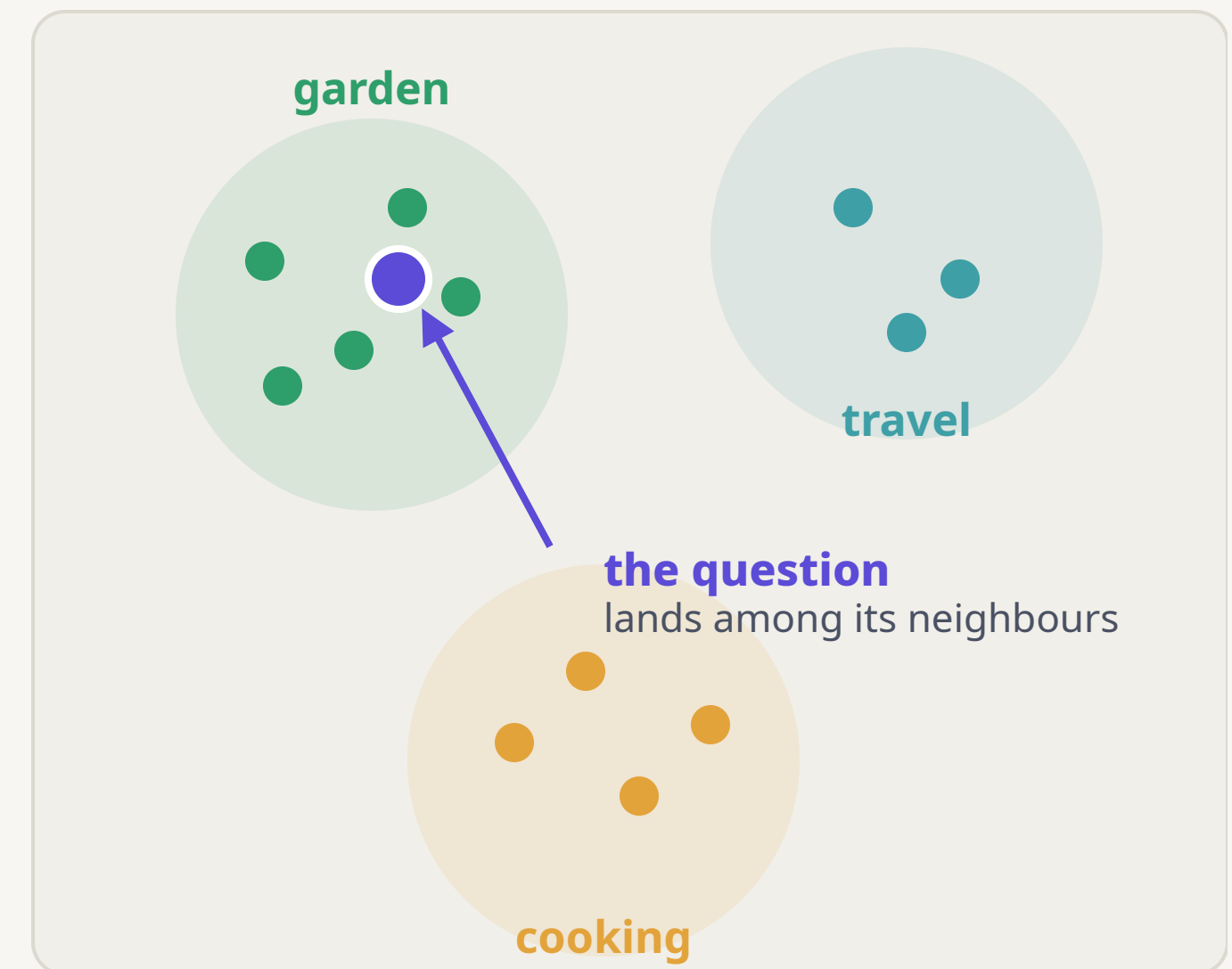
embedding
model

A vector

- [0.12, -0.80, 0.33, ...]
- 1,536 numbers: its place on a map of meaning

Why a place on a map?

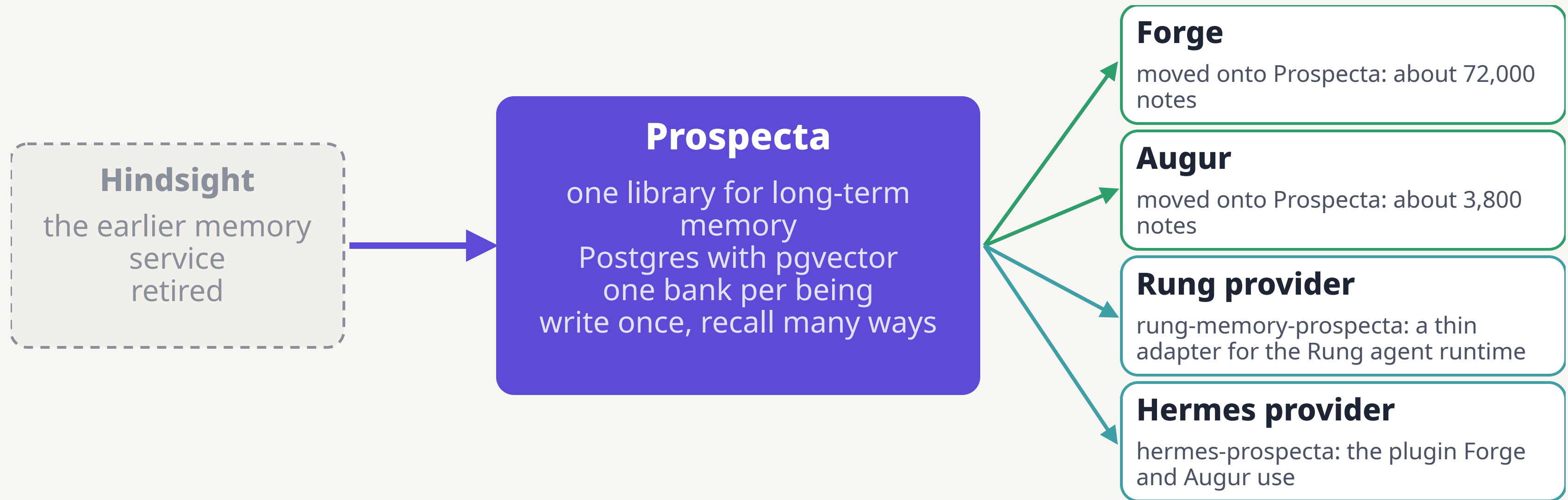
- Texts about the same thing land close together; unrelated texts land far apart.
- Searching is then simple: put the question on the map and read off its nearest neighbours.
- The map is stored in Postgres, with the pgvector extension that finds the nearest points fast.



WHY IT EXISTS

One memory library replaced Hindsight

Forge and Augur moved onto it; Rung and Hermes plug in through thin providers.



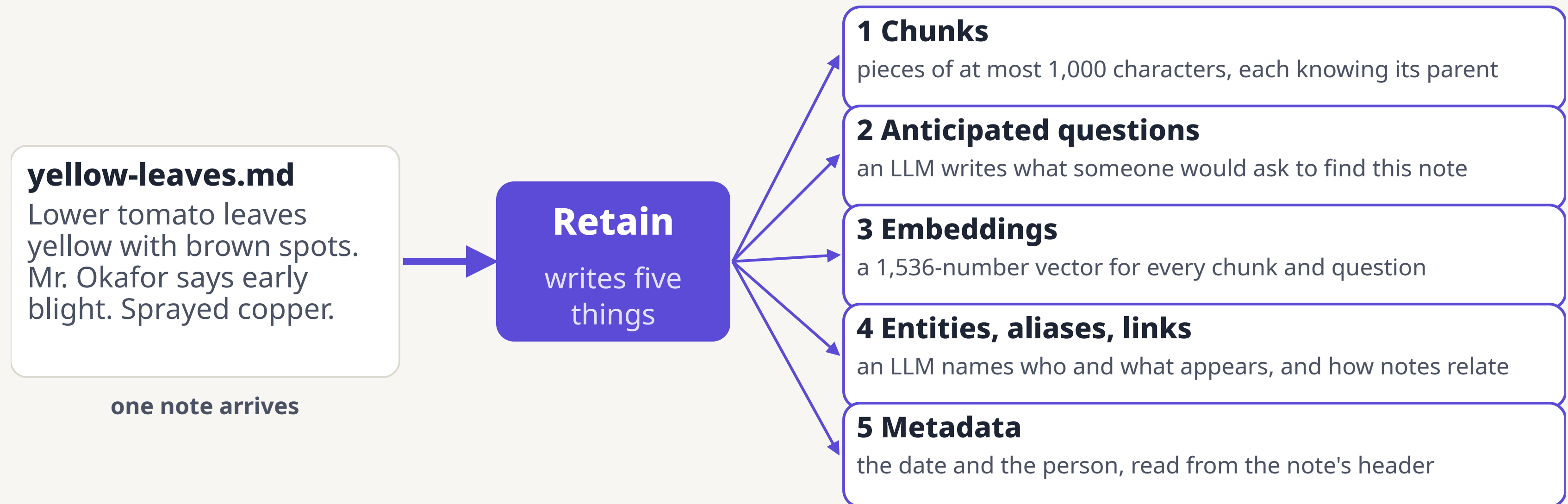
STATUS Moving the two live banks to the full hybrid recall is staged: dump, rehearse, shadow reads, then cut over.

How it works

What is written for every note, and the plan for reading.

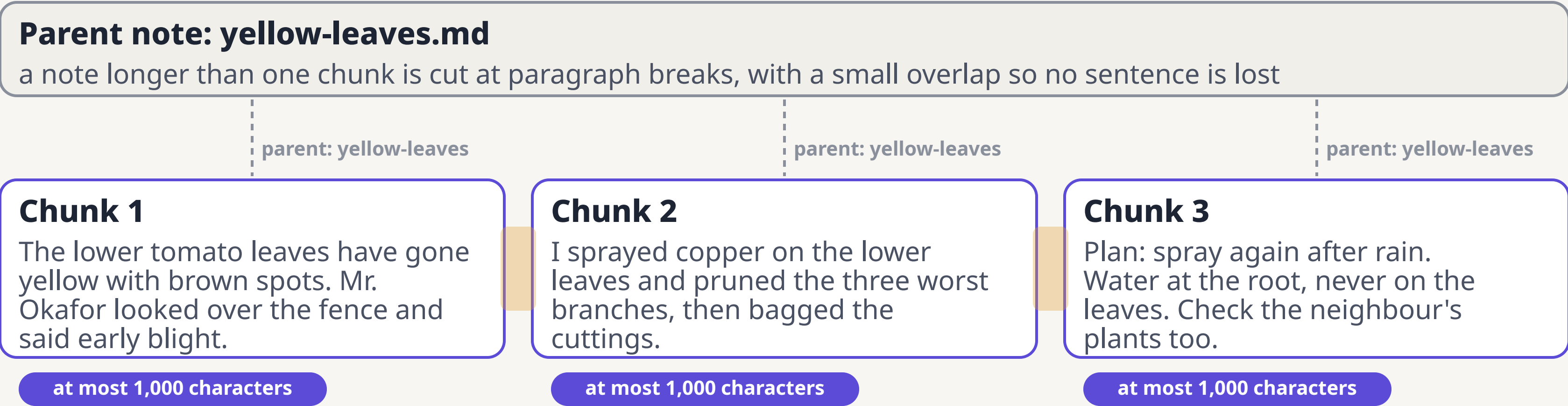
Five things are written for every note

The note is kept whole. Around it, Prospecta writes what makes it findable.



Chunks keep their parent note

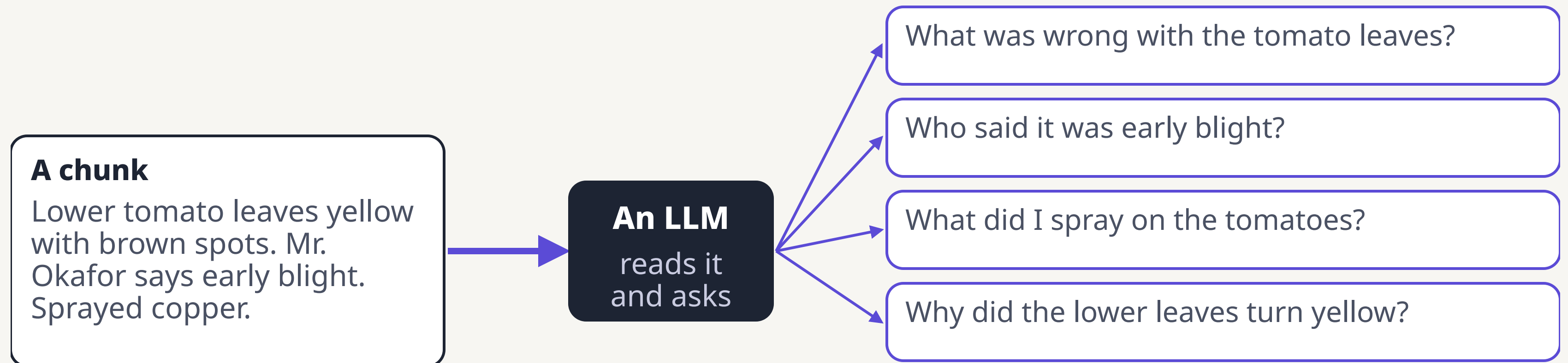
Right-sized pieces find precisely; the parent note gives the context.



The orange strips are the 100-character overlap shared by neighbouring chunks.
A search hits a chunk; the answer cites the parent note.

The writer guesses what will be asked

An LLM writes the questions each chunk could answer; they point back to it.

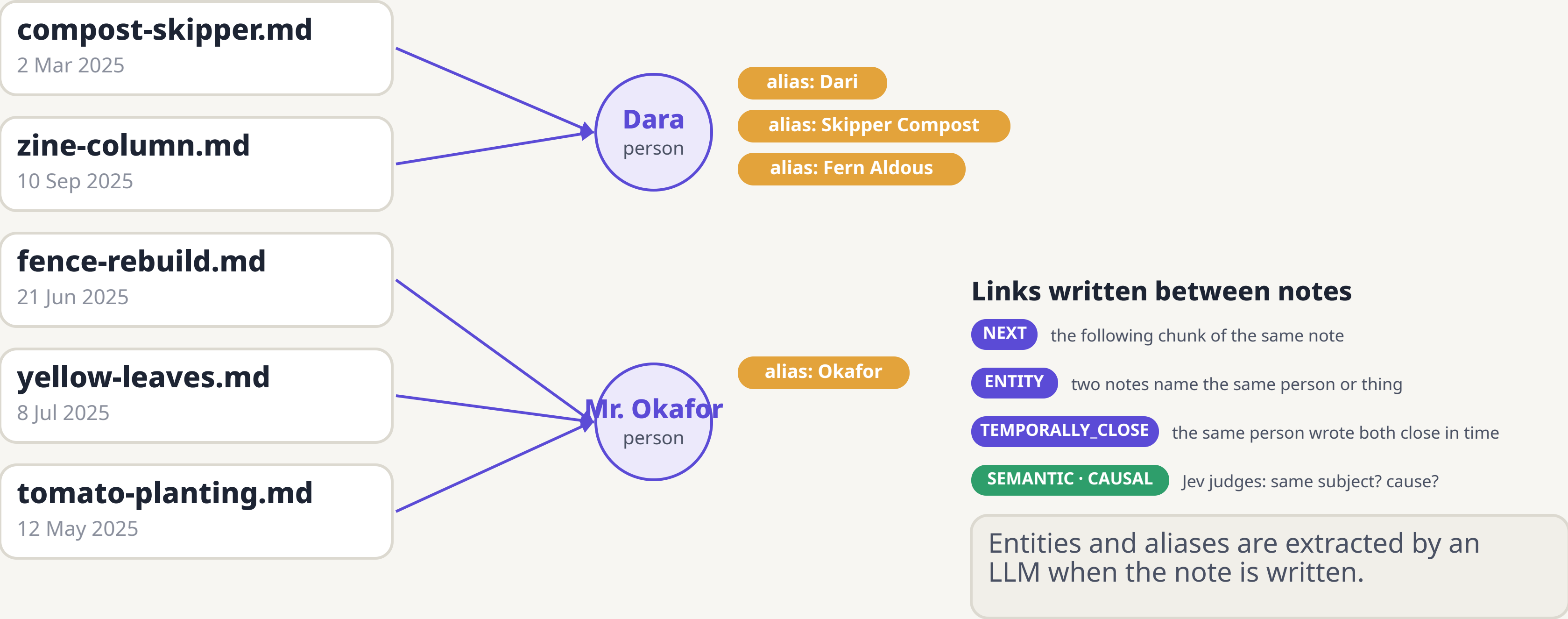


Each question is embedded too and points back to its chunk.

The reader's question will be compared with these.

Entities, aliases and links, written by an LLM

Who and what a note mentions, the other names they go by, and how notes relate.



RECALL

Recall is one plan with seven steps

Cheap searches first; language models only where they earn their cost.



Everything before step 5 is cheap

Search in Postgres costs nothing but a fraction of a second. The language models are what cost money and time.

Every step is recorded

What was asked, which channel found what, the reranker's grades and the answer are kept in Postgres tables, so any recall can be replayed.

Every technique

What each one is, what it is good and bad at, shown on the diary.

Dense vector search over right-sized chunks

Embed the question; return the chunks whose vectors sit nearest.

Question: when did the dark-fruited heirloom vines go into the ground?



No shared words with the note, yet its point is the nearest.

Good at

- Meaning: finds the note even when no word matches.
- Paraphrases, loose descriptions, questions.

Weak at

- Exact names, numbers, codes: close is not the same as equal.
- Dates and who: a note from the wrong month can sit nearby.

MEASURED

1,000-character chunks with a strong 1,536-d embedder: hit@10 0.92, against 0.72 for a small model. A title prefix lowered it to 0.86.

True BM25: search that weighs rare words

Count the question's words in each chunk, weighting rare words up and long chunks down.

Question: Okafor thermometer, as the notes' own words

word	notes holding it	weight (rarer is higher)
the	9 of 9	0.06
Okafor	3 of 9	1.05
thermometer	1 of 9	1.90

Rare words weigh more
A word in one note out of nine says a lot. A word in every note says nothing.

Repeats saturate
Saying a word ten times counts only a little more than saying it three times.

Short notes count more
A hit in a short note is stronger than the same hit lost in a long one.

Good at

- Exact names, numbers, codes and rare words.
- Needs no model at all: free and instant.

Weak at

- Paraphrase: a question with none of the note's words finds nothing.
- Common words mislead it: the word 'first' can pull the wrong note.

MEASURED

BM25 alone: hit@10 0.76; 1.00 on precise lookups, 0.00 on paraphrase. Postgres full-text rank needed every word and matched 1 question in 100.

The anticipated-question channel

Match the reader's question against the questions written when the notes were retained.

The question

Why did my tomatoes get sick?

Anticipated questions written at retain time, compared with it

What was wrong with the tomato leaves?	0.81 → yellow-leaves.md
What did I spray on the tomatoes?	0.66 → yellow-leaves.md
When were the tomatoes planted?	0.58 → tomato-planting.md
How many kilos did the harvest give?	0.31 → first-harvest.md

Good at

- The reader's words differ from the writer's.
- Costs nothing at question time: the work was done when the note was written.

Weak at

- The LLM may anticipate the wrong questions: alone it reaches 0.74 hit@10 against 0.92 for chunks.
- Needs an LLM call for every note written.

MEASURED Alone it reaches 0.74 hit@10. Removing it from the full blend costs 0.08 hit@1 and 0.04 hit@10.

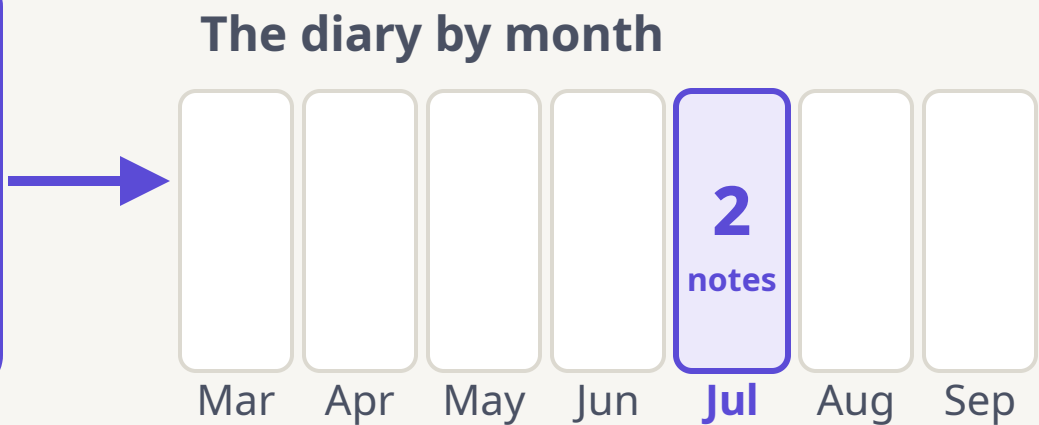
Metadata filters and scope promotion

Read people and dates out of the question; boost the notes that fit, never remove any.

QUESTION What went wrong in the garden in July 2025?

Plan reads it

- people: none
- date_from: 2025-07-01
- date_to: 2025-07-31
- hard: yes



Scope, never exclusion

The notes inside the filter get a score boost; no note is ever removed. Two of 19 extractions would have thrown away their own answer if used as a hard filter. Promotion is off by default.

Good at

- Who and when: scoped questions 0.70 to 1.00 hit@10.
- Free: a column lookup in Postgres.

Weak at

- Only as good as the extraction: a wrong date or name misleads.
- Needs metadata: notes without a date or person cannot be scoped.

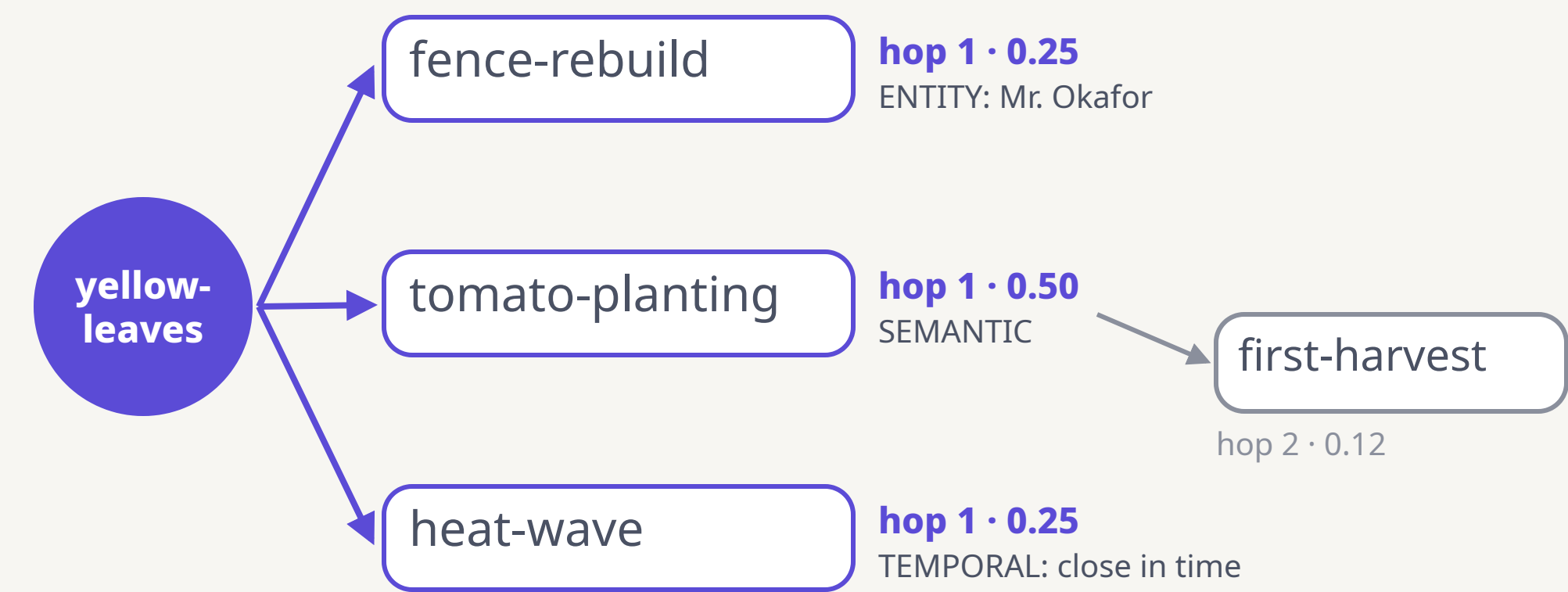
MEASURED

Scoped questions: 0.70 hit@10 for dense search alone, 1.00 with the metadata channel. The promotion re-sort is off by default.

The Jev-Mem graph channel

Start from found notes and walk entity and similarity links, two hops, fading with distance.

Seed: yellow-leaves, found by the other channels (score 1.00)



score = seed score x decay 0.5 x link-type weight x link confidence. Weights: SEMANTIC 1.0, CAUSAL 0.8, TEMPORAL 0.5, ENTITY 0.5.

Good at

- Finds neighbours no word or meaning match reaches: the same person, an earlier cause.
- Cheap to run: about 0.2 seconds in the latest build.

Weak at

- Measured gain: 0.00 to 0.01 hit@10. It found a note nothing else found for 2 of 264.
- Costs an LLM at write time; a hub person links to hundreds of notes.

MEASURED Removing the graph changes hit@10 by 0.00 to 0.01. It was the only finder of a gold note for 2 of 264, and it is built, kept, and cheap.

Jev: a cheap judge of relevance

Jev answers a 0 to 3 question per candidate. It reorders; it never invents a candidate.

QUESTION What did Mr. Okafor say was wrong with the tomatoes?

yellow-leaves.md



3

Holds the answer

tomato-planting.md



1

Same subject only

lisbon-stay.md



0

Unrelated

Jev, a small model made to judge relevance, answers a 0 to 3 question per candidate. About \$0.0003 a call.

Where Jev works here

- judges which notes to link at write time
- scores candidates in the reranker (hit@1 0.73 alone)
- gates Sonnet: a top score of 2.95 or more skips it

Why not use Jev for everything

- It reorders what others found; it never invents a candidate.
- As the only reranker it is 40 times cheaper than Sonnet but reaches 0.73 hit@1 against 0.87.

Fusion: weighted reciprocal rank fusion

Each channel votes by rank; a vote of rank r from a channel of weight w is worth $w / (60 + r)$.

note	dense rank in each channel	BM25	questions	equal weights	weights 4, 1, 1
tomato-planting	#1	#3	#2	0.0484	0.0976
seed-order	#4	#1	#1	0.0484	0.0953
first-harvest	#2	#2	-	0.0323	0.0806
yellow-leaves	#3	-	-	0.0159	0.0635
fence-rebuild	-	-	#3	0.0159	0.0159

The weights

- dense chunks 4
- metadata 3
- BM25 1
- questions 1
- graph 1

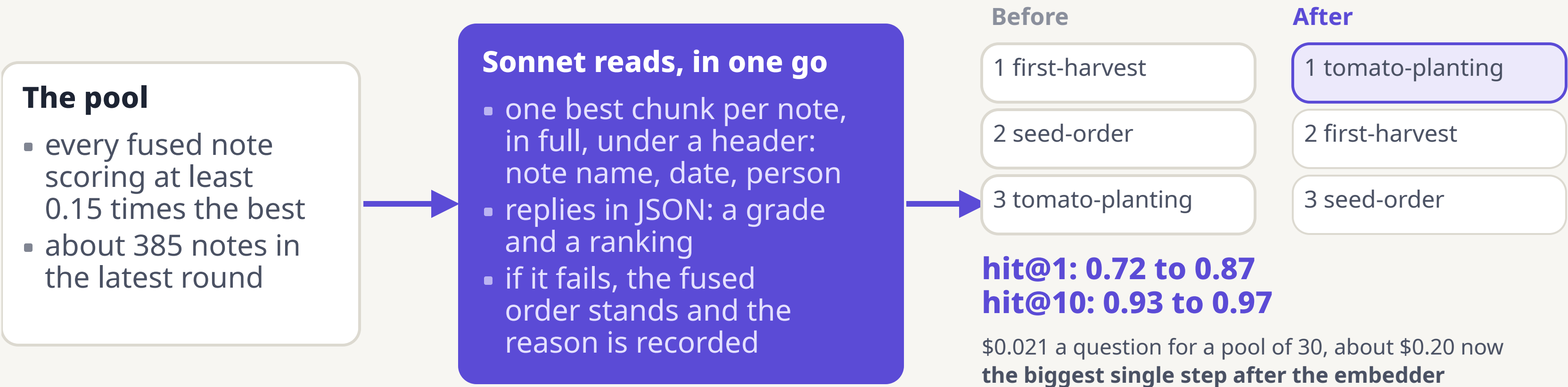
score = sum over channels of weight / (60 + rank). A rank counts for less the lower it sits.
Equal weights: seed-order and tomato-planting nearly tie, and seed-order wins. Weighted: tomato-planting wins.

k = 60

WHY WEIGHTS Equal weights: hit@1 0.59. Weights fitted on the 100 questions: 0.72, at the same hit@10 of about 0.93.

Re-ranking: an LLM reads and reorders

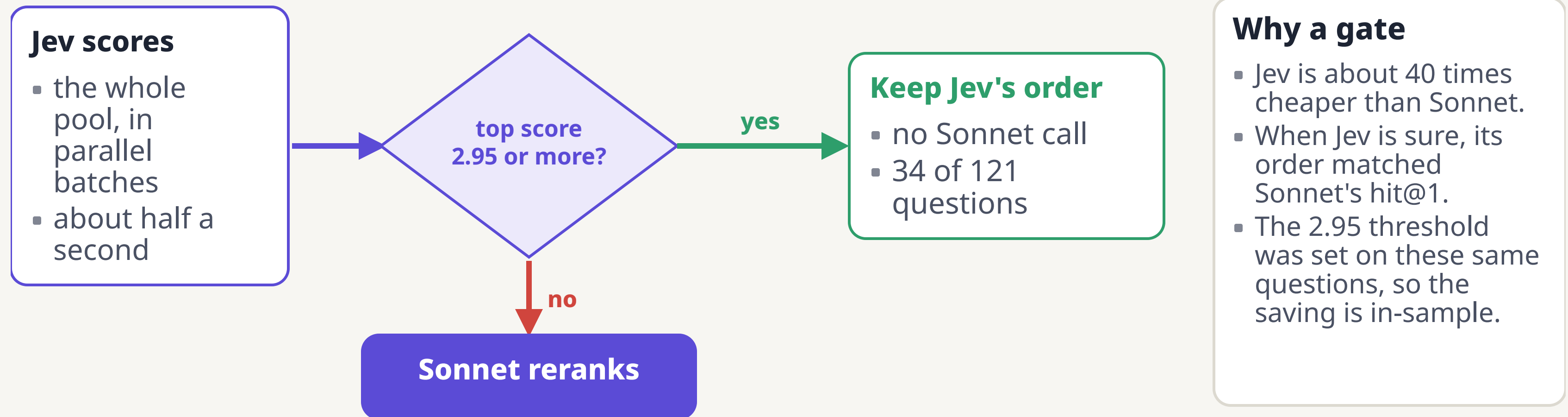
Sonnet reads the best chunk of every pooled note and returns a grade and a ranking.



MEASURED The biggest step after the embedder: hit@1 from 0.72 to 0.87, and paraphrase hit@10 from 0.75 to 0.95.

The Jev gate: skip the costly step when sure

Jev scores the pool first. A top score of 2.95 or more means Jev's order stands.



The 0.7/0.3 blend: tried, then turned off

Blend the reranker's order with the fused order. It hurt, so it is off by default.

The blend: $\text{score} = 0.7 \times \text{reranker position} + 0.3 \times \text{fused score}$

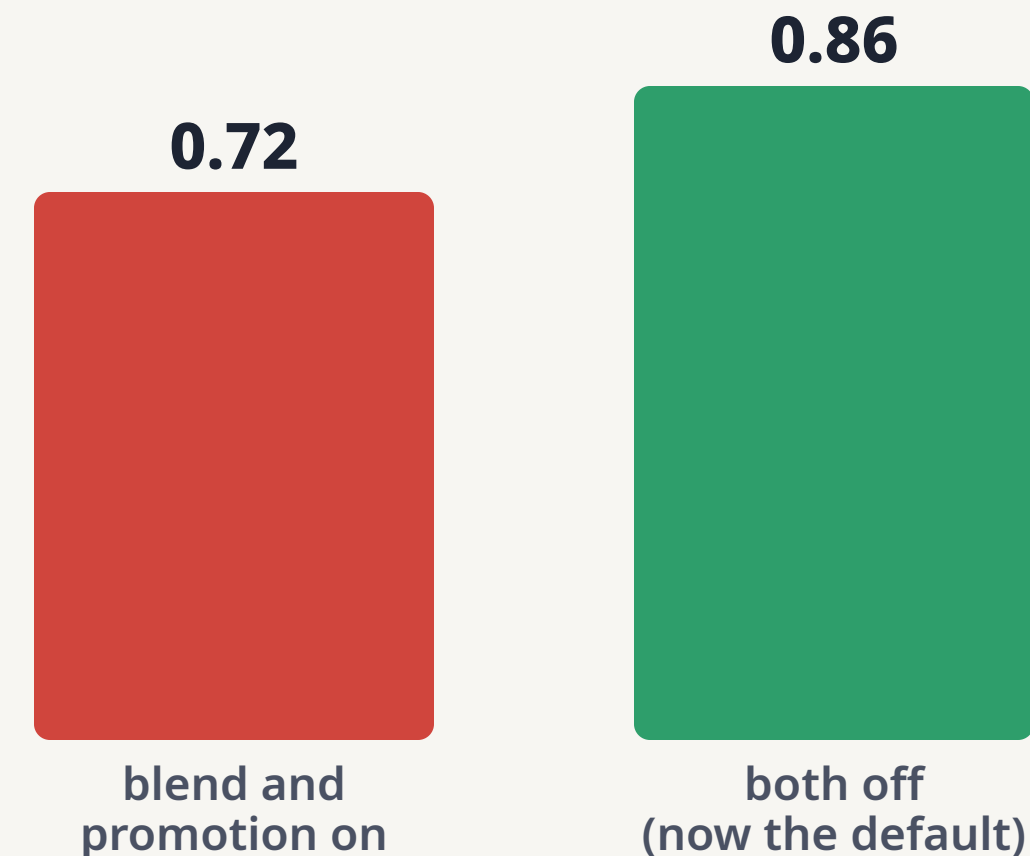
hit@1 on 120 questions

Tuned for a pool of 30

- The reranker's first and second choice differ by $0.7 / 29 = 0.024$ per place.
- Enough to matter against the fused term.

Run on a pool of about 385

- The same step is $0.7 / 384 = 0.002$.
- The fused score now decides, and the reranker only breaks ties.
- Switched off by default; the code stays for tests.

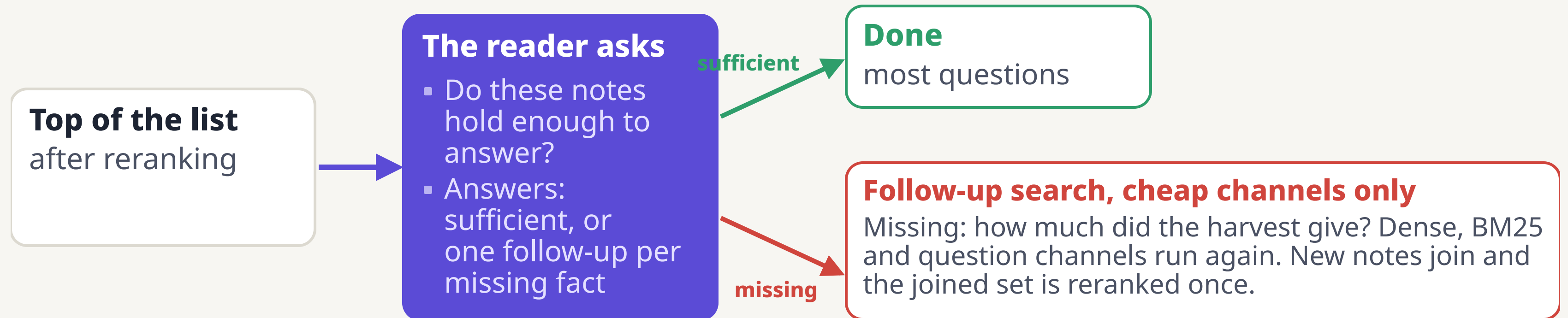


THE LESSON

A tuned constant is a hidden assumption. When the pool grew from 30 notes to about 385, the blend flattened the reranker's signal.

The second hop: look again when evidence is thin

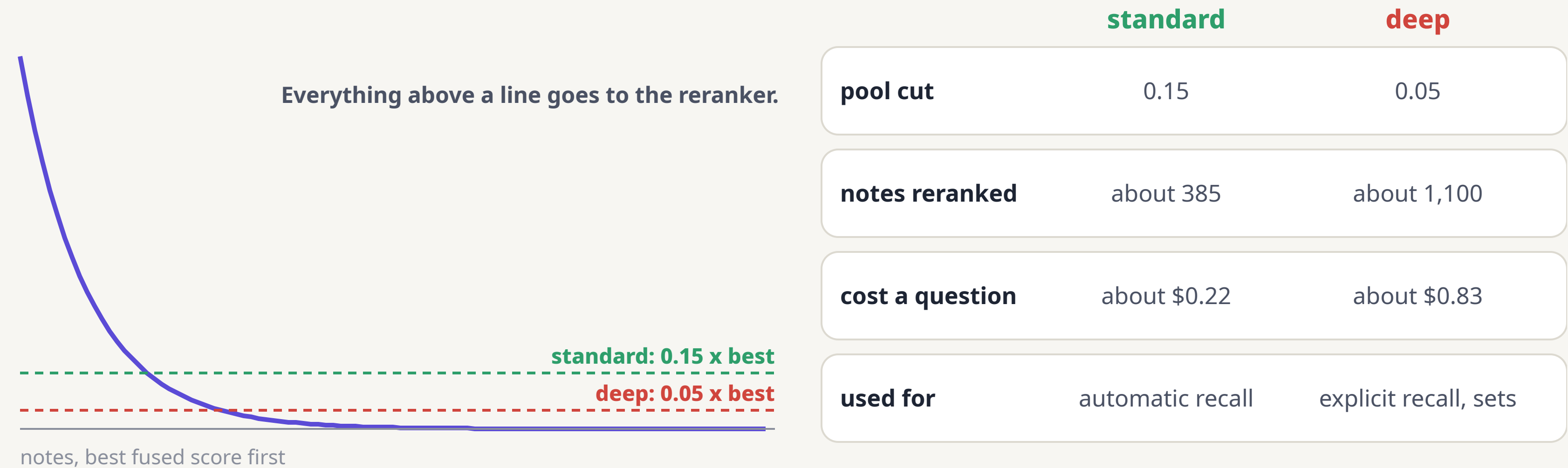
A reader checks whether the notes can answer; if a fact is missing it searches once more.



At most one hop. In the latest rounds the reader asked for a follow-up on 9 of 121 ordinary questions and 5 of 21 set questions; the hop lifted hit@1 by about 0.03 and set cover by 0.04.

Depth: standard or deep

Depth names how wide a net goes to the reranker: 0.15 or 0.05 of the best fused score.



MEASURED Deep costs about four times as much and gave no extra gain on set questions: set cover fell from 0.74 to 0.68.

Synthesis: a grounded, cited answer

The model answers only from the evidence handed over and cites each claim by note name.

QUESTION How did the tomatoes do?

Evidence handed over, whole

first-harvest.md

First big harvest: nine kilos of Black Krim. Sauce: garlic, basil, red...

yellow-leaves.md

Lower tomato leaves yellow with brown spots. Mr. Okafor says early bli...

heat-wave.md

37 degrees. Watered twice. The basil bolted....

**Sonnet
writes**
from the
evidence
only
cites every
claim

Answer

Nine kilos of Black Krim came in on 14 August [first-harvest]. Trouble started in July: early blight on the lower leaves, treated with copper [yellow-leaves], and a heat wave that bolted the basil [heat-wave].

If the evidence lacks it, the answer is 'not in memory'. A cited name that is not in the evidence is flagged.

MEASURED Judged correct on 82 of 100 questions with the full pipeline, against 44 to 60 for any single technique.

Map-reduce for set questions

When the answer is a set, read every related note in batches, then merge.

QUESTION What are Dara's nicknames?

1 Resolve

Dara is an entity, with aliases Dari, Skipper Compost, Fern Aldous

2 Gather

- every note tied to Dara, not just the top ten
- read in batches of eight

3 Map

- each batch: pull out the asked facts, each with its note
- Dari [compost-skipper]
- Fern Aldous [zine-column]

4 Reduce

- merge, drop repeats, keep citations
- Dari, Skipper Compost, and the pen name Fern Aldous

Why not a top-ten list?

A set has no 'best ten'. Search ranks by closeness, so a note that names a nickname without the word 'nickname' sits far down.

On the real notes

A question like this found all 5 names with citations, for \$0.42 and 98 seconds. Map-reduce costs a model call per batch.

The name resolver: Okafor finds Mr. Okafor

Names match by their core words, not by exact spelling; aliases lead to the person.

A name matches when every core word of the shorter name appears in the longer

you search	stored as		
Okafor	Mr. Okafor	match	'Mr.' is dropped, one word left matches
okafor, mr	Mr. Okafor	match	order and case do not matter
Oka	Mr. Okafor	match	a 3-letter start of a word matches
Dari	Dara	match	an alias row points to its person
Ofak	Mr. Okafor	no match	letters out of order: no match

On the real notes

- A question about one person, stored as 'Mr. <surname>', first found 0 candidate notes.
- With the resolver: 22 notes, all 4 answers among them.

Scoring

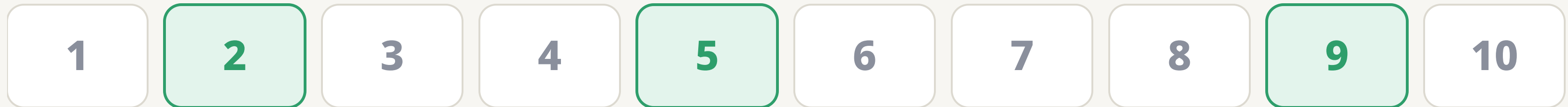
How a score is built, and what it can and cannot tell us.

HOW A SCORE IS BUILT

hit@1, hit@10 and cover@10

Each question has gold notes: the right answers. A score asks where they land in the top ten.

One question, the top ten notes returned. Gold notes (the right answers) are green.



A fourth gold note sits at place 14: outside the top ten.

hit@1

0

- Is a gold note first?
- Here: no. Place 1 is not gold.

hit@10

1

- Is any gold note in the top ten?
- Here: yes. Places 2, 5 and 9.

cover@10

0.75

- What share of all gold notes is in the top ten?
- Here: 3 of 4 gold notes.

A TINY WORKED EXAMPLE

Four questions, three scores

Averages over the questions give the numbers in this deck.

question	gold at places	hit@1	hit@10	cover@10
A When were the tomatoes planted?	1	1	1	1.00
B What went wrong in July?	4	0	1	1.00
C Everything about Mr. Okafor	2, 5, 9, 14	0	1	0.75
D Which kale did I order?	23	0	0	0.00
Average of the four		0.25	0.75	0.69

hit@1 is 1 of 4. hit@10 is 3 of 4. cover@10 is (1 + 1 + 0.75 + 0) / 4.

A set question (C) has several gold notes, so only cover@10 shows how much of the set came back.

With 100 questions, one question is 0.01: read only differences of 0.03 or more.

What the test questions were, honestly

Model-written

Sonnet wrote 100 questions from the notes. The paraphrase class shares no word with its note on purpose, harsher on word search than real use.

Two golds

Gold 1: the notes each question was written from. Gold 2: any note that holds the answer, judged by Sonnet.

The owner's own

The headline set holds one question of the owner's own so far. The harness takes more, appended as they come.

Small sample

With 100 questions one question is 0.01. Weights and the gate's 2.95 were fitted on the same questions.

Same model family

The judge and the reranker are the same family, which flatters gold 2. Gold 1 is the independent check.

Real notes, never quoted

1,845 real diary notes were used. None is quoted here; the examples are invented.

READ AS differences under 0.03 at hit@10 and 0.05 at hit@1 are noise at this size.

Four kinds of question

A lookup, a paraphrase, a dated question and a set, each on the diary.

A precise lookup

Rare words in the question: word search and meaning search both find it.

QUESTION What temperature was the soil when I planted the Black Krim tomatoes?

Dense weight 4	BM25 weight 1	Questions weight 1	Metadata weight 3	Graph weight 1
1 tomato-planting	1 tomato-planting	1 seed-order	no person or date in the question: abstains	1 fence-rebuild
2 seed-order	2 seed-order	2 tomato-planting		2 yellow-leaves
3 first-harvest	3 first-harvest	3 yellow-leaves		neighbours of the seeds
4 yellow-leaves	Black Krim, soil, planted			

meaning matches

↓ fuse, then rerank

Result

1 tomato-planting	2 seed-order	3 first-harvest
-------------------	--------------	-----------------

Every channel agrees. BM25 and dense both nail it because the question carries the note's own rare words. The answer: 16 degrees [tomato-planting]. Hit at place 1.

A paraphrase

No shared word: word search is lost, meaning and questions carry it.

QUESTION When did the dark-fruited heirloom vines first go into the ground?

Dense
weight 4

1 seed-order

2 first-harvest

3 tomato-planting

4 compost-skipper

BM25
weight 1

1 first-harvest

2 zine-column

only 'first' matches: wrong notes

Questions
weight 1

1 tomato-planting

2 seed-order

'When were the tomatoes planted?'

Metadata
weight 3

nothing to scope on: abstains

Graph
weight 1

1 fence-rebuild

meaning: right area, gold third

↓ fuse, then rerank

Result

1 tomato-planting

2 seed-order

3 first-harvest

No word is shared, so BM25 is lost and even leads astray. Dense and the question channel carry it, and the Sonnet reranker reads for meaning and puts the gold note first. Paraphrases went from 0.75 fused to 0.95 reranked, then 1.00 after the hop.

A time-scoped question

July 2025 is a date range: the metadata channel returns the notes written then.

QUESTION What went wrong in the garden in July 2025?

Dense weight 4	BM25 weight 1	Questions weight 1	Metadata weight 3	Graph weight 1
1 yellow-leaves	1 fence-rebuild	1 yellow-leaves	1 yellow-leaves	1 tomato-planting
2 first-harvest	2 compost-skipper	2 heat-wave	2 heat-wave	
3 tomato-planting	no word says July: noise		the two July notes	
4 heat-wave				

meaning, but July is just a word

↓ fuse, then rerank

Result

1 yellow-leaves	2 heat-wave	3 first-harvest
-----------------	-------------	-----------------

The plan reads 'July 2025' as a date range and the metadata channel returns exactly the notes written then. Its weight of 3 lifts both July notes above the August harvest that dense search liked. Scoped questions: 0.70 for dense alone, 1.00 after the filter.

A set question

Several notes, one name: only name-aware channels see them all.

QUESTION What are Dara's nicknames?

Denseweight 4

1 compost-skipper

2 fence-rebuild

3 lisbon-stay

4 seed-order

BM25weight 1

1 compost-skipper

2 zine-column

both notes name Dara

Questionsweight 1

1 compost-skipper

2 heat-wave

Metadataweight 3

Dara is not the diary's person: abstains

Graphweight 1

1 zine-column

2 compost-skipper

entity Dara and its aliases

the word 'nickname' pulls one note

↓ fuse, then rerank

Result

1 compost-skipper

2 zine-column

3 fence-rebuild

BM25 and the graph carry this one, because only those see the name Dara. The pen name note never says 'nickname', so a top-ten list can bury it. Sets go to map-reduce: read every note tied to Dara. Set cover@10 rose from 0.60 to 0.76 over the rounds.

Which technique carries which question

No single channel wins every row; the full pipeline is at or near the top of all four.

	Dense	BM25	Questions	Metadata	Graph	Full pipeline
Precise lookup hit@10	0.90	1.00	0.70	0.45	0.10	1.00
Paraphrase hit@10	0.65	0.00	0.60	0.10	0.30	1.00
Time-scoped hit@10	0.70	0.90	0.60	1.00	0.10	0.90
SET question cover@10	0.39	0.68	0.23	0.00	0.27	0.67

Outlined: the best single channel in the row. Source: evaluation round 2, 120 questions.

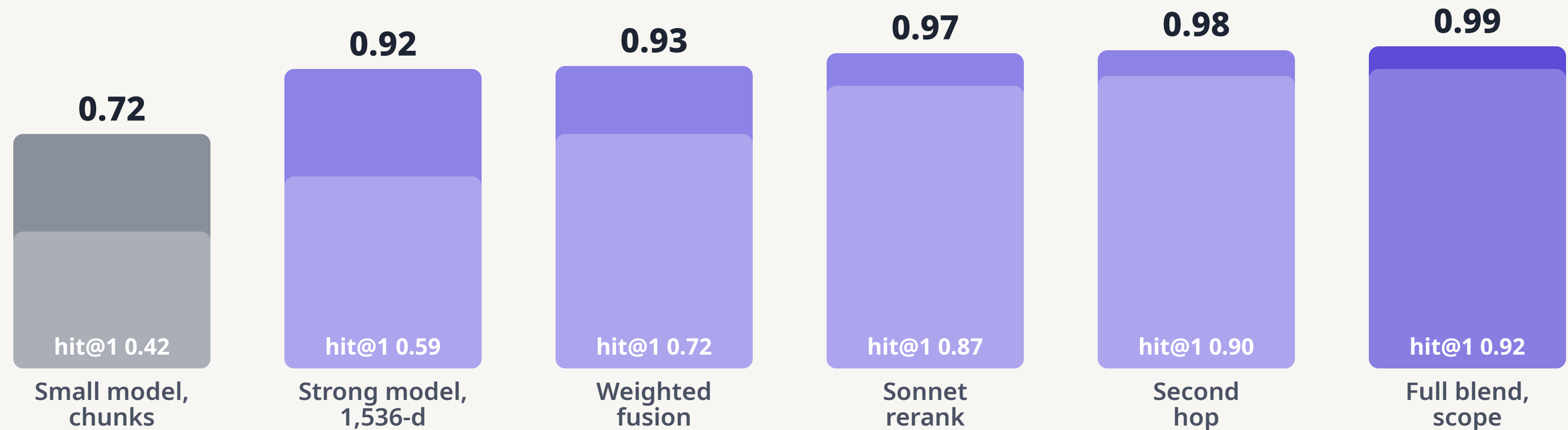
Why hybrid beats any single technique

The ladder, step by step, from our own measurements.

THE LADDER

The ladder: 0.72 to 0.99 hit@10

Each step adds one technique to the last; the answer's place rises with every rung.



hit@10 on 100 questions (strict gold). The lighter part of each bar is hit@1.

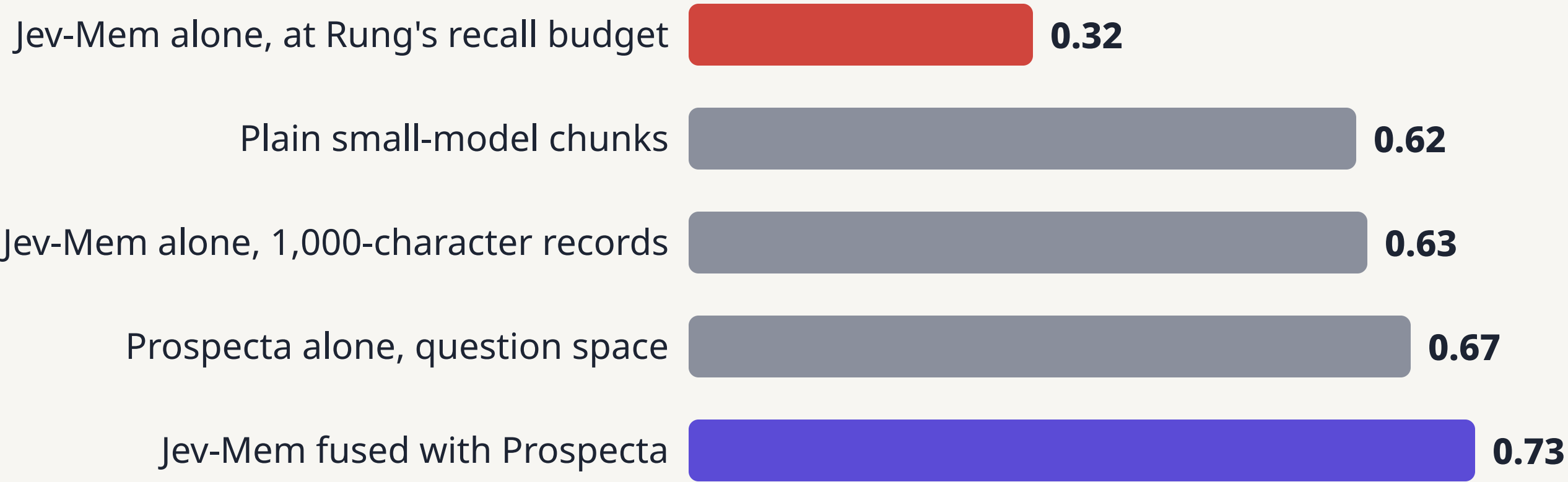
READ AS no step is magic; the embedder and the reranker are the two big ones, and the last rungs fix the rare misses.

WHERE WE STARTED

No single arm passed 0.67

The first test, on 60 questions: each technique against the others.

The first test: hit@10 on 60 questions, one technique against another



No single arm passed 0.67. A hybrid of two reached 0.73; the strong embedder and the stages did the rest.

Tradeoffs

Cost, latency, write-time cost, and what we learned the hard way.

COST PER RECALL

Cost: the reranker is the bill

Reading hundreds of notes is the price of the accuracy.

Cost of one recall, in US dollars

rerank Jev gate reader plan, embed

Standard recall



Set question



Deep recall



The reranking model is almost all of it: it reads hundreds of notes per question.

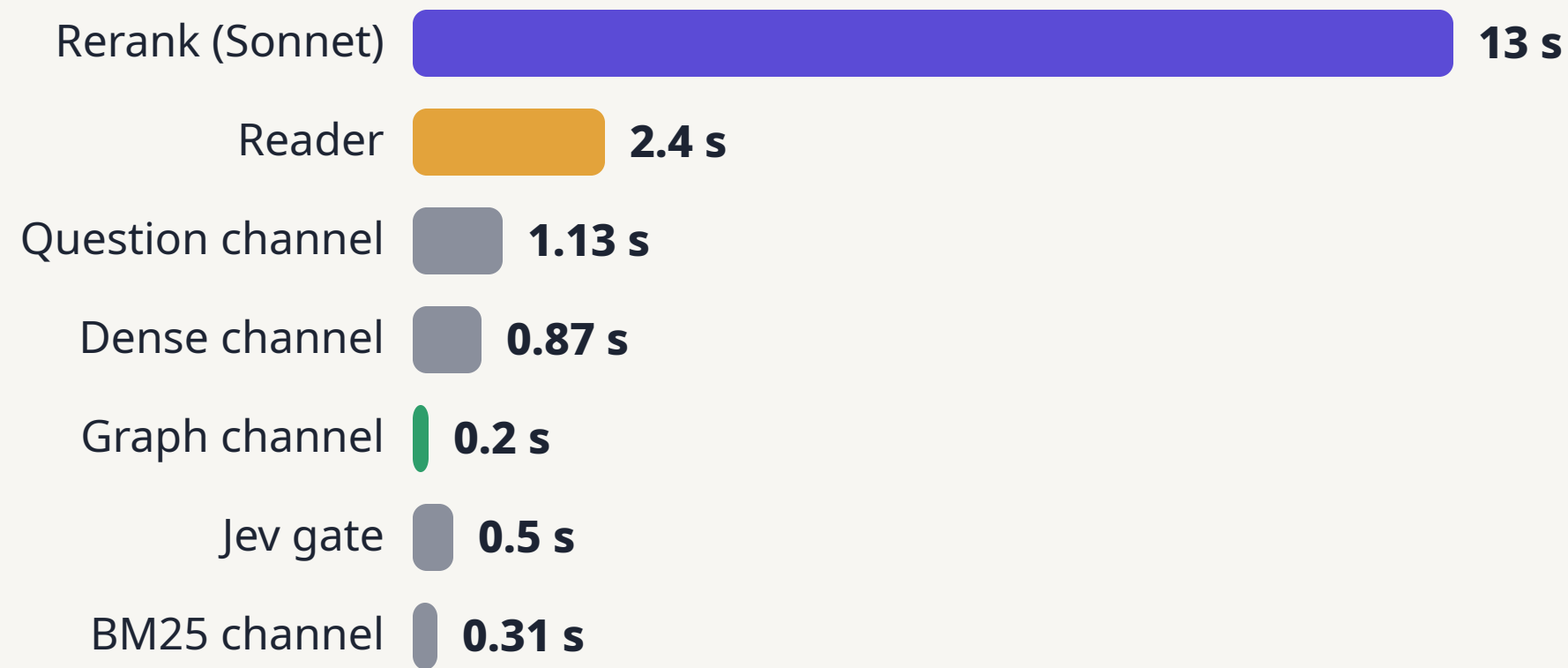
Deep reads a pool of 1,000 to 1,200 notes and costs about four times as much. On set questions it gave no extra gain: cover@10 fell from 0.74 to 0.68, more notes but a worse ranking. Sets go to map-reduce instead.

LATENCY

Latency: about 19 seconds at the median

Language-model stages dominate; the searches take a second.

Mean seconds in each stage of a standard recall



p50 19 s

Half finish in 19 seconds or less; the slowest tenth take 35 or more.

18% over 30 s

22 of 121 recalls ran past the 30-second default timeout.

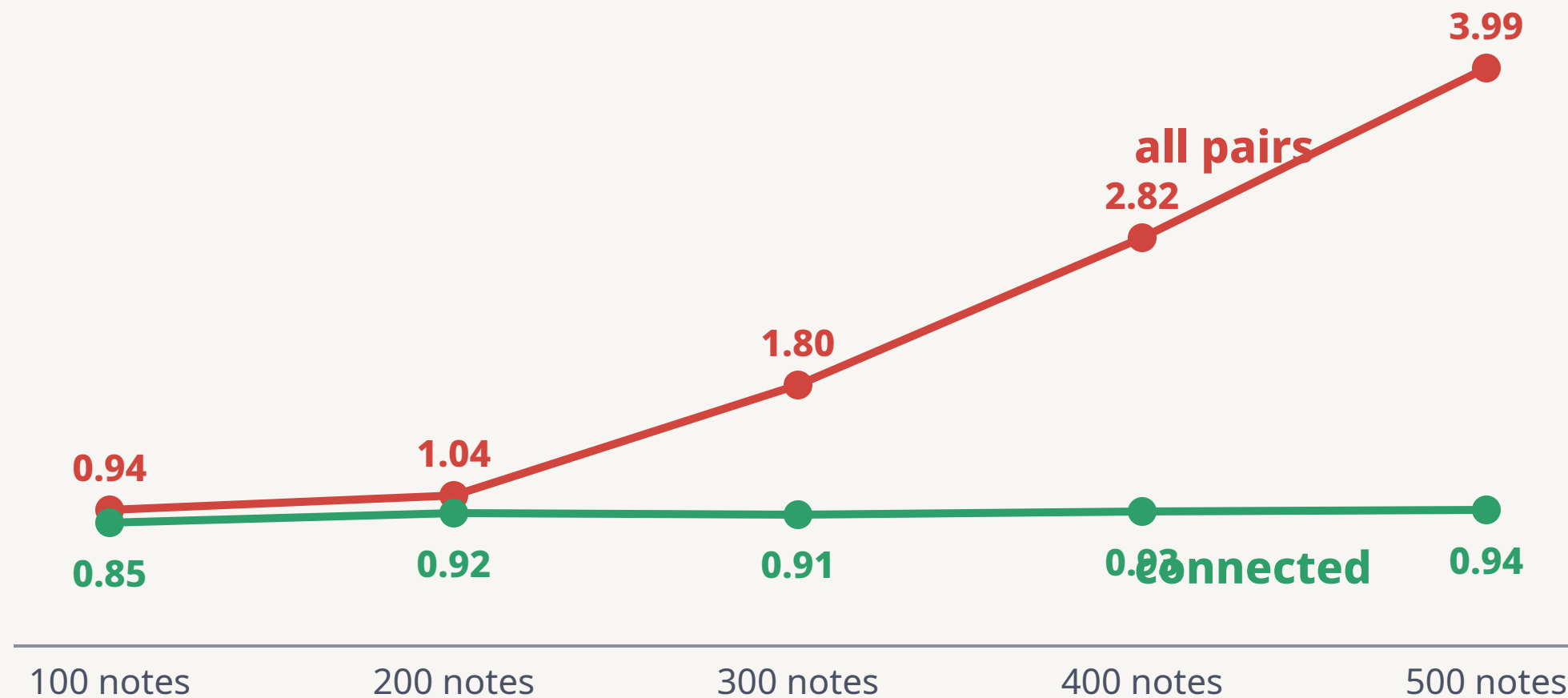
Graph: 0.2 s p50

0.47 s at the slow tenth, after a rewrite; it once took 40 s.

Write-time cost: the linker and the import

Linking every close pair is thorough and grows with the bank; connected linking stays flat.

Jev calls per note as the bank grows (real 500-note import)



Write-time cost, estimated

- Forge: about \$41 for the linker plus about \$128 for extraction, about \$170 in all.
- Augur: \$2 to \$10 for the linker.

The two modes

- All pairs judges every close pair of chunks: the default, quality over speed.
- Connected judges far fewer: 0.94 calls a note, 7.7 times cheaper, 11.6 times fewer semantic links.

What went wrong, and what it taught

A blend that hurt

Blend and promotion re-sort cut hit@1 from 0.86 to 0.72. Both are now off by default.

A quadratic linker

An early cut made Jev calls grow with the square of the bank: a projected \$29 to \$37 import against \$3.60.

A dead lexical channel

Postgres full-text needed every word and matched 1 question in 100. True BM25 replaced it.

A slow graph

Removing its caps made it 40 seconds a question. A rewrite brought it to 0.2 seconds.

No caps on content

No limit may clip content or evidence. A limit stays only if a provider forces it, and the full text is logged.

Earn your weight

Graph, entity links and Jev-Mem native recall moved scores 0.00 to 0.02, so channels earn weight by ablation.

RULE measure each change on the same questions before it ships, and keep the switch.

Areas for improvement

What is still weak, said plainly.

Where recall is still weak

Sets are not solved

Set cover@10 is 0.76, and all gold notes are in the top ten for 10 of 20 set questions.

Map-reduce precision

Hub people pull 1,248 to 1,621 notes, 68 to 88 percent of the bank, for 6 of 21 questions.

Standard misses names

In the latest round the standard top ten held 2 of the 5 names; deep held all 5.

Latency

Median 19 seconds; 18 percent past the 30-second default timeout.

The graph is unproven

No measurable gain yet; the multi-hop questions that would test it are not in the set.

Filter extraction

About 3 percent of calls return an empty reply and fall back to a plain pattern match.

Where the evidence and the operations are thin

The question set

Model-written, one question of the owner's. More real questions and a second judge family would settle it.

Unmeasured thresholds

The pool, reader and hop cuts and the gate's 2.95 are our starting choices, tuned in-sample or on synthetic data.

Forge's bank

Not yet converted: it needs a link pass of about \$170 and a long run, and several steps are unverified on live data.

Concurrency

An early bug under eight writers is fixed; keep the safety-net pass that re-links failures.

English only, blocking

Stemming is English; every call blocks until done.

Cost control

Rung declares a per-recall budget, and a deep recall can pass it.

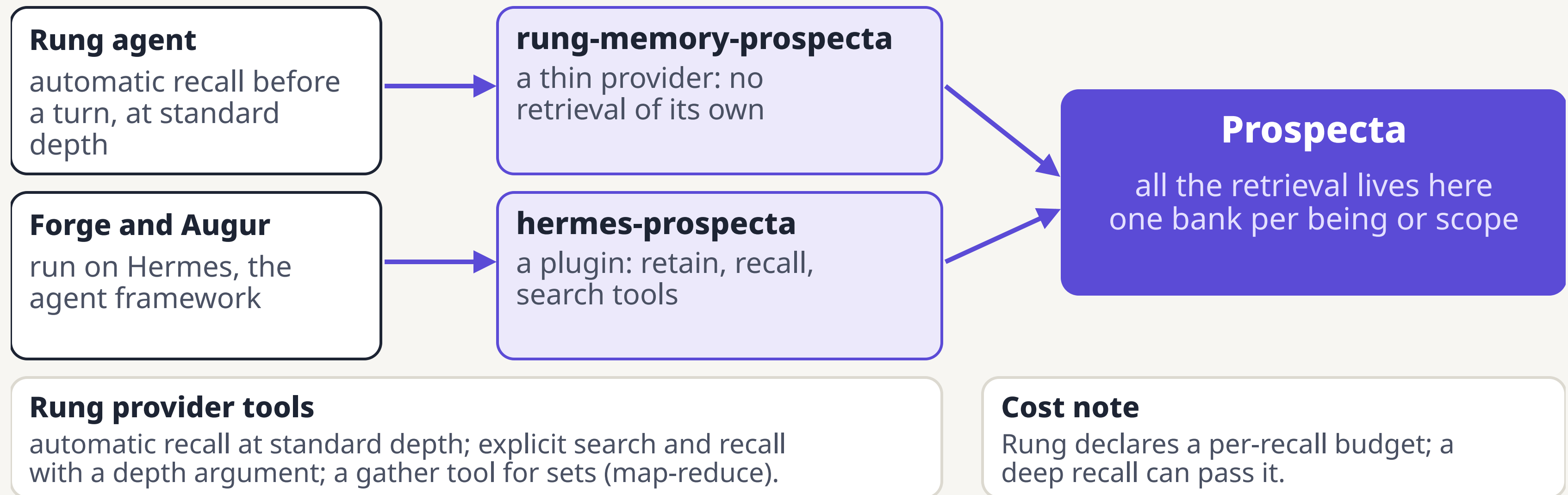
Plugging in

Rung and Hermes reach Prospecta through thin providers.

PROVIDERS

Rung and Hermes plug in through thin providers

The providers hold no retrieval logic; Prospecta does the work.



IN CLOSING

Retain what happened, recall what matters

One bank per being, on Postgres: hybrid recall beats any single technique, at a cost we can name.

0.72 to 0.99

hit@10 across the ladder, from a small model over chunks to the full blend, on 100 questions.

\$0.22

for a standard recall, almost all of it the reranker; about \$0.38 for a set question.

0.76

set cover@10 after the name resolver: the gap we still have to close.